Lingling Chen

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RESEARCH INTERESTS

My research focuses on embodied intelligence, with an emphasis on the reliability and generalization of vision-language-action (VLA) models for robotic manipulation, including uncertainty estimation and object-centric reasoning. I am interested in reasoning and generalization in real-world robotic environments, as well as combining agent planning with VLA models.

EDUCATION

Ph.D. Student in Computer Science, The University of Texas at Dallas Aug. 2026 – Present
- Advisor: Prof. William J. Beksi, Physical Reasoning and Intelligent Systems (PRIS) Lab

M.S. in Computer Science, The University of Texas at Arlington 2024 – 2025
- GPA: 4.0/4.0
- Thesis: An Uncertainty-Guided and Failure-Aware Vision-Language-Action Framework for Robotic Control

B.S. in Software Engineering, Yangzhou University 2014 – 2018

PUBLICATIONS

[1] Zongyao Lyu, **Lingling Chen**, and William J. Beksi. "SentryVLA: Uncertainty-Guided Action Selection and State Monitoring for Vision-Language-Action Robotic Control." *Accepted to the 10th Conference on Robot Learning (CoRL 2026).* [Project Page]

[2] **Lingling Chen**, Zongyao Lyu, and William J. Beksi. "RIGOR-VLA: Object-Centric Role Grounding with Relation-Aware Target Selection for Vision-Language-Action Models." *Under review.*

RESEARCH EXPERIENCE

Research Assistant Aug. 2026 – Present
PRIS Lab, The University of Texas at Dallas
- Investigated object-centered target reasoning based on spatial relations to improve VLA policy generalization to novel objects, layouts, and tasks.
- Studied VLA reliability for fine-manipulation tasks by building a UR5 simulation environment in Isaac Sim, synthesizing execution data with variations in backgrounds, object positions, lighting, and camera perturbations from collected data, and developing a simulation evaluation environment.

Research Assistant 2024 – Jul. 2026
Robotic Vision Laboratory, The University of Texas at Arlington
- Developed uncertainty-aware action-selection and state-monitoring methods for reliable VLA-based robotic control.
- Evaluated pretrained VLA policies in LIBERO simulation and on a physical UR5 robot arm without modifying the underlying policy architecture.

SELECTED RESEARCH PROJECTS

Waves-Workcell: Robotic Assembly and Peg-in-Hole Insertion 2026

- Built a UR5e assembly system that localizes 3D-printed parts, performs grasp-and-transport motions, and completes tight-tolerance insertion.
- Combined simulation, demonstration-finetuned $\pi_{0.5}$ VLA policies, multi-camera calibrated perception, and planning modules for skill orchestration and failure recovery.
- Synthesized execution data from real robot collection across varied backgrounds, object positions, lighting, and camera perturbations, and built a simulation evaluation environment to assess VLA generalization and reliability in closed-loop simulation.
- Achieved 76.7% success (46/60 episodes) with the $\pi_{0.5}$ model in closed-loop simulation evaluation and 75% success (15/20 episodes) in real excution.

RIGOR-VLA: Object-Centric Role Grounding for VLA Models 2026 – Present

- Developed an object-centric framework that assigns execution, target, and placement roles to instance-aligned object representations for instruction-sensitive manipulation.
- Designed relation-aware geometric reasoning to distinguish target instances when multiple objects share the same noun, reducing wrong-object failures.
- Built the LIBERO-RIGOR benchmark to evaluate target selection in seen, unseen, and counterfactual layouts.

Demogen_Libero: Data Generation for LIBERO-RIGOR 2026

- Generated and filtered retargeted demonstration data across new object layouts for LIBERO-RIGOR, with object-centric RGB-D, segmentation, pose, and bounding-box annotations.

SentryVLA: Uncertainty-Guided Action Selection and State Monitoring 2025 – 2026

- Developed a lightweight runtime reliability framework for frozen VLA policies without retraining or changing model architecture.
- Used conformal quantile regression to rank candidate actions by calibrated deviation from expert behavior.
- Designed a Mahalanobis-distance state monitor to detect anomalous execution contexts from policy latent states.

PROFESSIONAL EXPERIENCE

Senior Software Test Engineer Jul. 2019 – Jan. 2023

ZKH Industrial Supply, Shanghai, China

- Built Python-based automation frameworks for large-scale e-commerce and supply-chain systems.

TECHNICAL SKILLS

- Programming Languages: Python, SQL, Java, C
- Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers
- Robotics and Embodied AI: Vision-Language-Action (VLA) models, LIBERO, Isaac Sim, UR5/UR5e, ROS2, OpenCV
- Research Methods: uncertainty estimation, conformal prediction, object-centric learning, failure detection, policy evaluation